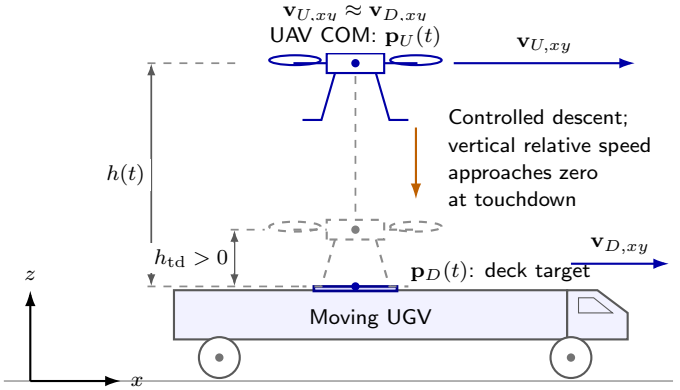
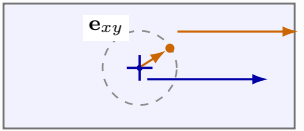
Side view: level deck with straight motion

Track the moving deck while descending



Top view: horizontal alignment



Cross: deck target; dot: UAV.
Arrows: horizontal velocity.

Permit descent only if

$$\|\mathbf{e}_{xy}\| \le \varepsilon_p$$

$$\|\mathbf{v}_{U,xy} - \mathbf{v}_{D,xy}\| \le \varepsilon_v$$

Reliable pose; clear deck.

U: UAV COM; D: deck target.

World frame: z up $\quad h(t) \downarrow h_{\mathrm{td}}$, the COM clearance at contact.